KT Decision Analysis (472)

			A: Music (2)		B: Art (3)		C: History (3)		D: Art 203 (3)		E: Geolg101(3)		F: Music205(2)	
Musts	1.		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO	
	2.		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO	
	3.		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO	
	4.		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO		**GO** / **NO** GO	
Wants		Weight	Rating	Score	Rating	Score	Rating	Score	Rating	Score	Rating	Score	Rating	Score
1-														
2-														
3-														
			TOTAL		**TOTAL**		**TOTAL**		**TOTAL**		**TOTAL**		**TOTAL**	

ADVERSE CONSEQUENCES TABLE

CONSEQUENCE	ADVERSE CONSEQUENCES			
	Probability of Occurrence	Seriousness	P * S	Total
SOLUTION NAME ( )				
A1				
A2				
A3				
SOLUTION NAME ( )				
B1				
B2				
B3				
SOLUTION NAME ( )				
C1				
C2				
C3				
SOLUTION NAME ( )				
D1				
D2				
D3				
SOLUTION NAME ( )				
E1				
E2				
E3				

Process Check

(team discuss, fill out and submit at Instructor's table)

	AGREE				DISAGREE
Today's class Helped Us Understand					
what Decision Analysis is	**1**	**2**	**3**	**4**	**5**
what MUSTS and WANTS are, and	**1**	**2**	**3**	**4**	**5**
what adverse consequences are.	**1**	**2**	**3**	**4**	**5**
The Team Feels <u>Very</u> Comfortable					
creating KT DA Tables,	**1**	**2**	**3**	**4**	**5**
creating Adverse Consequences Tables,	**1**	**2**	**3**	**4**	**5**
using KT DA Tables, and	**1**	**2**	**3**	**4**	**5**
using Adverse Consequences Tables.	**1**	**2**	**3**	**4**	**5**
The team stayed focused on task.	**1**	**2**	**3**	**4**	**5**

List a **change** in **today's** class that **would enable** your **team** to **learn more** effectively:

Exxon Valdez Problem

It is 12:45 AM in the morning, March 24, 1989; you have just been alerted that the Exxon Valdez tanker has run aground on the Bligh Reef and is spilling oil at an enormous rate. By the time you arrive at the spill, 6 million gallons of oil have been lost and the oil slick extends well over a square mile. A meeting with the emergency response team is called. At the meetings it is suggested that a second tanker be dispatched to remove the remaining oil from the Exxon Valdez. However, the number of damaged compartments from which oil is leaking is not known at this time and there is concern that if the tanker slips off the reef, it could capsize if the oil is only removed from the compartments on the damaged side.

The use of chemical dispersants (i.e., soap-like substances) that would break up the oil into drops and cause it to sink is suggested. However, it is not known if there is sufficient chemical available for a spill of this magnitude. The marine biologist at the meeting objected to the use of dispersants, stating that once these chemicals are in the water, they would be taken up by the fish and thus they would be extremely detrimental to the fishing industry. The use of floatable booms to surround and contain the oil also brought about a heated discussion. Because of the spill size, there is not enough boom material even to begin to surround the slick. The Alaskan governor's office says the available material should be used to surround the shore of a small village on a nearby island.

The Coast Guard argues that the slick is not moving in that direction and should be used to contain or channel the slick movement in the fjord. The Department of Wildlife says the first priority is the four fisheries that must be protected by the boom or the fishing industry will be depressed for years, perhaps generations to come. A related issue is that millions of fish were

scheduled to be released from the fisheries into the oil-contaminated fjord two weeks from now. Other suggestions as to where to place the boom material were also put forth at the meeting.

The Historical story



THE Exxon Valdez

Exxon Valdez was the original name of an oil tanker owned by the former Exxon Corporation. It gained widespread infamy after the March 24, 1989 oil spill in which the tanker hit Prince William Sound's Bligh Reef and spilled an estimated 11 million gallons of crude oil.

The vessel had an all steel construction, built by National Steel and Shipbuilding Company in San Diego. A relatively new tanker at the time of the spill, it was delivered to Exxon in December 1986. The tanker was 300 m long, 50 m wide, and 27 m in

depth (987 ft by 166 ft by 88 ft), weighing 30,000 tons empty and powered by a 31,650 shp (23.60 MW) diesel engine. The ship could transport a maximum of 1.48 million barrels (200,000 t) at a sustained speed of 16.25 knots (30 km/h) and was employed to transport crude oil from the Alyeska consortium's pipeline terminal in Valdez, Alaska, to the lower 48 states of the United States. The vessel was carrying about 1.26 million barrels, or about 53 million gallons, when it struck the reef. The accident would have been contained, but the captain only reported it as a minor accident.

After the spill, the Exxon Valdez was towed to San Diego, arriving on July 10 and repairs began onboard on June 30, 1989. Approximately 1,600 tons of steel were removed and replaced Cheese in the month of July 1989, the tanker left harbor after $30 million of repairs.

Team: **Session C.5.2** **Date: / /202**

	Situation Analysis: Exxon Valdez			
Situations / Problems / Projects	**Timing** (H,M,L)	**Trend** (H,M, L)	**Impact** (H,M, L)	**Process** (PA ,DA, or PPA)
1.				
2.				
3.				
4.				
5.				
6.				
7.				
8.				
9.				
10.				

Process Check
Team: session C-5-

(team discuss, fill out and submit at Instructor's table)

	AGREE				DISAGREE
Today's Class Helped Us Understand					
what a Situation Analysis is	**1**	**2**	**3**	**4**	**5**
The Team Feels Very Comfortable					
creating KT SA Tables,	**1**	**2**	**3**	**4**	**5**
The team stayed focused on task.	**1**	**2**	**3**	**4**	**5**

List a **change** in **today's** class that **would enable** your **team** to **learn more** effectively:

Solar Car Race - Problem Definition

The Race

The cars must rely on the sun for all their power. On days when there is little sunlight, the cars are powered by a battery pack. The battery pack must be charged by the sun.

The basic rules of the race include:

- **The race lasts for seven days**
- **The car's are required to drive a leg of the race each day, with a time penalty imposed if they do not finish.**
- **The cars are allowed to race from 9:00 a.m. until 6:30 p.m.**
- **The cars are allowed to charge their battery pack from the time they finish racing for the day until 8:30 p.m.**

The Team

The solar car is surrounded by two vans that serve to protect the delicate vehicle. Additional responsibilities include:

- Lead Van - Navigates course and makes sure traffic laws are followed.
- Chase Van - Determines optimum speed for the car and handles communication between all vehicles.
- Some teams also have a van that is several hours ahead of the convoy to relay weather information.

The Strategy

The main strategic aspect of the race is determining the optimum speed for the car, which is based on:

- Amount of sunlight
- Charge remaining in the batteries
- Weather forecast

If the speed is set too high, the batteries might run out of power and, depending on the amount of sunlight, the car might not be able to finish a leg of the race. Likewise, if the speed is set too low, the car will lose time to its competitors and not have as much time at the end of the day to charge its batteries.

It is essential to test the car to determine the relationship between power use and velocity. This way, the strategist can compute the speed of the car, based on its available power.

The Car :

- A large, delicate solar array panel collects the sunlight.
- The solar array can be used to power the car of charge the battery pack
- The tires are similar to a mountain bike's, only modified so that they are thinner.
- The frame, which is the part of the car that supports the body, is susceptible to cracking. Cracks in the frame or loose fasteners on the suspension can cause the chassis to fail and the car to detach from the wheels.
- A loose connection in one of the wires in the car can cause the motor controller, a device that controls the timing of the car's electric motor, to fail.
- The battery pack is comprised of many individual batteries. During pre-race testing, information can be obtained about individual batteries, but, during the race, the strategist can only obtain information about the performance of the battery pack as a whole.

	KT Potential Problem Analysis			
Potential Problems	**Consequences**	**Possible Causes**	**Preventive Actions**	**Contingency Plan**
A.				
B.				
C.				
D.				
E.				
F.				
G.				
H.				
I.				

Process Check

Team: session C-5-2

(team discuss, fill out and submit at Instructor's table)

	AGREE				DISAGREE
Today's Class Helped Us Understand					
what Potential Problem Analysis is	1	2	3	4	5
what a Preventive Action is, and	1	2	3	4	5
what Contingency Actions are.	1	2	3	4	5
The Team Feels <u>Very</u> Comfortable					
creating KT PPA Tables,	1	2	3	4	5
The team stayed focused on task.	1	2	3	4	5

List a **change** in **today's** class that **would enable** your **team** to **learn more** effectively: